

1. GitHub Repository Setup:

- **Task:**
 - Create a new GitHub repository named "CloudDevOpsProject"
 - Initialize the repository with a README.
- **Deliverables:**
 - URL to the GitHub repository.

2. Containerization with Docker:

- **Task:**
 - Deliver docker-compose.yml file for building and testing the application with the Database locally.
 - Find the source code from: <https://github.com/Ibrahim-Adel15/iVolveFinalProject.git>
- **Deliverables:**
 - Docker-compose file is committed to the repository.

3. Infrastructure Provisioning with Terraform:

- **Task:**
 - Deliver Terraform modules for AWS resource provisioning:
 1. Network Module (VPC, public & private subnets, IGW, NAT, and Network ACL)
 2. Server Module (EC2 instances for Jenkins with Security Groups)
 3. EKS Module (EKS with 2 worker nodes in different private subnets & Availability zones)
 4. ECR Module
 - Use S3 Terraform Backend state.
- **Deliverables:**
 - Terraform modules are committed to the repository.

4. Configuration Management with Ansible:

- **Task:**
 - Deliver Ansible playbooks for Jenkins EC2 instance configuration:
 - Install Java.
 - Install Jenkins.
 - Install the required packages (Docker, Trivy).
 - Use Ansible roles.
 - Use Dynamic Inventory.
- **Deliverables:**
 - Ansible inventory, playbook, and roles are committed to the repository.

5. Container Orchestration with kubernetes:

- **Task:**
 - Create iVolve namespace.
 - Configure Deployment for each microservice.
 - Configure Service for each microservice.
 - Configure Statefulset for database.
 - Configure Headless service for Statefulset.
 - Configure StorgaeClass to store data from the Statefulset.
 - Configure ConfigMap and Secret for the environment variables.
 - Create ingress for accessing the frontend microservice.
- **Deliverables:**
 - Required YAML files are committed to the repository.

6. Continuous Integration with Jenkins:

- **Task:**
 - Deliver Jenkins pipeline for each microservice.
 - Stages: Build Image, Scan Image, Push Image, Delete Image Locally, Update Manifests, Push Manifests.
 - Use Shared Library.
- **Deliverables:**
 - Jenkins files are committed to the repository.
 - Shared library directory (vars) is committed to the repository.

7. Continuous Deployment with ArgoCD:

- **Task:**
 - Configure ArgoCD to sync and deploy your app into cluster.
- **Deliverables:**
 - ArgoCD Application is committed to the repository.

8. Documentation:

- **Task:**
 - Deliver comprehensive documentation:
 - Setup instructions.
 - Architecture overview.